

The one-dimensional Taylor tree to all orders, monomial perturbation

Claude, with Daniel Murfet

2026-09-06

Abstract

The general-order version of units 28–29: for $\xi(u) = \xi_0 + cu^m$ and every truncation order N ,

$$\left| Z_{\text{chart}} - \sum_{p < N} \frac{(\beta c)^p}{p!} \frac{1}{2k} n^{-\mu_p} S_{\mu_p + p/2}(\xi_0) \right| \leq \frac{(\beta|c|)^N}{N!} \frac{1}{2k} n^{-\mu_N} S_{\mu_N + N/2}(\xi_0 + |c|b^m) + \sum_{p < N} \frac{(\beta|c|)^p}{p!} \text{tail}_p, \quad \mu_p = \frac{h+mp+1}{2k},$$

with the tails exponentially small. This is `thm:TaylorTree/cor:standardintegralexp` to every order in one dimension: the exponents μ_p run along the arithmetic progression $(h+1)/(2k) + (m/2k)\mathbb{N} \subseteq \Lambda(h, k)$ and every coefficient is an explicit fluctuation function. The analytic input is a global remainder bound $|e^x - \sum_{p < N} x^p/p!| \leq \frac{|x|^N}{N!} e^{|x|}$. Zero `sorry/axiom`.

1 Setting

Unit 28 handled $N = 2$ by hand. The same decomposition $e^x = \sum_{p < N} x^p/p! + R_N(x)$ with $x = \beta\sqrt{n}u^kcu^m$ works at every order: the finite Taylor part is a sum of insertion integrals (each known exactly up to a tail by units 22/24/25), and the remainder is dominated pointwise by $\frac{|x|^N}{N!}e^{|x|}$, which on the chart is $\frac{(\beta|c|)^N}{N!}$ times the N -th insertion integrand at the shifted argument $\xi_0 + |c|b^m$ — so the remainder integral is again an insertion value, at order $\mu_N + N/2$.

2 The things formalised

```
theorem abs_exp_sub_sum_range_le (x : ) (N : ) :
  |Real.exp x - ∑ p in Finset.range N, x ^ p / p.factorial|
  |x| ^ N / N.factorial * Real.exp |x|

theorem standardIntegral1D_taylor_tree ( n c b : ) (h k m N : )
  (h : 0 < ) (hm : 1 ≤ n) (hb : 0 < b) (hk : 0 < k) :
  |Z_chart - ∑ p in range N, ( c )^p/p! * ((1/(2k)) n^(-_p) S_{_p+p/2}( ))|
  ( |c|)^N/N! * ((1/(2k)) n^(-_N) S_{_N+N/2}( + |c| b^m))
  + ∑ p in range N, ( |c|)^p/p! * tail_p
```

3 Proof strategy

The exponential bound is the unit-28 series argument with the shift 2 replaced by N ; the only new ingredient is $N!p! \leq (p+N)!$, from `Nat.factorial_mul_factorial_dvd_factorial_add`.

The main theorem decomposes the integrand pointwise as $\sum_{p < N} G_p + g_R$ with $G_p(u) = \frac{(\beta c)^p}{p!} u^{h+mp} (\sqrt{n} u^k)^p e^{\text{pop}}(u)$ (the identity $G_p(u) = u^h e^{\text{pop}}(u) x^p / p!$ plus `Finset.mul_sum`), integrates term by term (`integral_finsetSum`, `integral_add`; all pieces continuous on the chart), rewrites each $\int G_p$ as insertion value minus tail exactly as in unit 26, bounds $|\int g_R|$ through the dominating N -th insertion at $\xi_0 + |c|b^m$, and closes with $|\sum| \leq \sum |\cdot|$ on the tails.

4 Roadblocks and resolutions

Essentially none — the file compiled at the first attempt apart from a deprecated name (`continuous_finset_sum` $\rightarrow$ `continuous_finsetSum`). The design choices that made this smooth were all inherited from units 26 and 28: opaque local definitions via `obtain g, hg : g, g = ...`, a universally quantified algebraic key : `SA ST IR, SA - ST + IR - SA = IR - ST` for the final rearrangement (robust to the goal's exact parenthesisation), and stating the remainder through the insertion lemma rather than by hand (no `rpow merge` needed).

5 What was Mathlib, what was new

Mathlib: the exponential series, `hasSum_nat_add_iff'`, `hasSum_le`, `Nat.factorial_mul_factorial_dvd_factorial_add`, `integral_finsetSum`, `continuous_finsetSum`, `Finset.abs_sum_le_sum_abs`. New: the global exponential Taylor remainder bound (a lean-common candidate alongside its $N = 2$ case) and the all-orders one-dimensional Taylor tree with explicit coefficients and remainder.

6 Lessons

Once the $N = 2$ case is written with opaque local definitions and the insertion identity as the only evaluator, the general- N proof is a near-verbatim generalisation; do the small case first only if it teaches the idioms, then generalise immediately rather than stacking special cases.

7 Follow-ups

The $O(n^{-\mu N})$ packaging (absorbing the N tails, each $n^{p/2} e^{-\varepsilon n}$) is the same routine as unit 29; the multivariate tree and analytic (non-monomial) ξ remain.